

Chapter 1. Financial Engineering and Financial Derivatives

Financial Engineering

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Content

1 Course Information

2 Financial Engineering

3 Financial Derivatives

- Futures and the Margin
- Forwards and Their Arbitrage Pricing
- Options

Course Information

- Course title: *Financial Engineering*, **FE** for short.
- Level: undergraduate, Grade 3 or 4.
- Prerequisite (先修课程): calculus; probability.

Relevant but **not required**: stochastic process; real analysis; measure theory; functional analysis.

- Textbook: slides (课件), uploaded on the BlackBoard platform.

Course Taught in English

- This course is delivered **completely in English**.
- All of the lecture language, slides, exams, assignments, and course notifications are demonstrated in English, while the important parts will have Chinese translations simultaneously.
- Students are **required** to finish the assignments and exams in English.
- Your rights:
 - ask the instructor to translate any part into Chinese;
 - ask problems in Chinese after class.

The Instructor



Dr. SHI, Chao (史钊). Email: shi.chao@sufe.edu.cn
Associate Professor (tenured) in SUFE.

Research area: financial derivative pricing, Malliavin calculus, financial risk management.

<http://simecv.sufe.edu.cn/page.aspx?id=72>

Lectures and Grading

- 2 credits, 32 class hours
- Please eat enough before class!
- Grading
 - usual performance 30%
 - open-book final exam 70%
- 11 quiz or assignments. (Why 11?)

Chapters

- Chapter 1. FE and Financial Derivatives
- Chapter 2. Option Pricing Theory (with Fundamental Stochastic Calculus)
- Chapter 3. Volatility Models
- Chapter 4. Numerical Methods in FE

General principles in the slides:

blue texts are important concepts or ideas;

red texts are core points of the entire course;

Any part with a tag “*” will not be involved in the exam.

Reference Books

- John Hull. *Options, Futures, and Other Derivatives*, 11ed.
 - “Wall Street Bible”
 - An encyclopedia (百科全书) of FE
 - Emphasize both theory and practice
 - Very high timeliness (时效性)
- Steven Shreve. *Stochastic Calculus for Finance II*.
 - Mathematical foundation of FE
 - In the framework of stochastic calculus

Definition of Financial Engineering

- Financial engineering is a multidisciplinary field involving financial theory, methods of engineering, tools of mathematics and the practice of programming.
 - MS in Financial Engineering, Columbia University
- Financial engineering draws on tools from applied mathematics, computer science, statistics and economic theory.
 - International Association of Financial Engineers, 2012

FE as a Discipline

- Financial engineering is an interdisciplinary, which studies financial markets and products with the aid of mathematics, primarily including probability, statistics, optimization, PDE, etc.
- Also called Mathematical Finance, Financial Mathematics, Computational Finance (with subtle difference in particular emphases).
- A young but promising discipline.
 - Since its advent in 1980s as a formal discipline, FE has been developing very quickly.
 - Provide deep insights into finance.

Main Academic Topics and Methods in FE

There are three main research domains in the area of FE.

- Portfolio selection
 - Optimization theory, control theory, behavior finance.
- Risk management
 - Measure theory, high-dimensional data analysis, hedging theory.
- Asset pricing
 - Stochastic calculus, dynamic modelling, PDE.

Portfolio Selection Theory

- Generally speaking, for one's investment, the higher the return, the higher the risk.
- Question: How to invest in a variety of derivative securities? Or equivalently, how to select a portfolio of derivative securities?
- No perfect answers yet, but FE has given deep insights.
 - Mean-variance analysis and Capital Asset Pricing Model (CAPM) (Markowitz, Nobel Prize in Economics in 1990)
 - Minimizing the variance with the same return level.

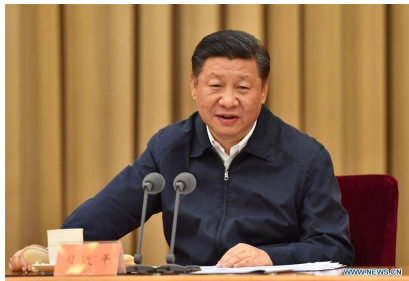
Financial Risks

- “Lehman Moment” and 2008 global financial crisis
- “816 乌龙指” event, \$85m fine on Everbright Securities, 2013
- Circuit breakers (熔断) for four times in U.S. stock market caused by the COVID-19 Pandemic, 2020
- Coal price regulation by NDRC (国家发改委), 2021
- RTS drops by 49% on the first day of Russia-Ukraine war, 2022
- China's Growing Local Government Debt, 2023

Systemic Risk and Systematic Risk

- Systemic risk (系统性风险) describes an event that can spark a major collapse in a specific industry or the broader economy.
- Systematic risk (系统风险) is the pervasive, far-reaching, perpetual market risk that reflects a variety of troubling factors.

President Xi's Speech



“防范化解金融风险，特别是防止发生系统性金融风险，是金融工作的根本性任务，也是金融工作的永恒主题。”

— President XI, Jinping on the 5th National Financial Work Conference, 2017

Asset Pricing Theory

- Basic Assets:
 - Stock, index, ETF
 - Commodity
 - Bonds, interest rate
 - Foreign exchange (FX, Forex, 外汇)
- Derivatives (introduced later in this chapter)
- The asset pricing theory investigate into the “fair” market price of an asset, under some given market conditions.

Wealth Management Product

- A **Wealth Management Product (WMP, 理财产品)** is a financial product sold by banks or other financial institutions. It typically provides higher interest rate than time deposit (定期存款).
- Example: “利得盈” WMP issued by CCB (中国建设银行) invests in fixed-income securities, which have low risk and high liquidity.
- Example: “汇得盈” WMP issued by CCB invests in various financial assets traded in international markets, such as FX and crude oil.

Structured Wealth Management Product

- “金钥匙如意组合” WMP issued by ABC (中国农业银行) provides a **random annualized rate of return** (年化收益率), which is determined by the performance of CSI 300 index (沪深300指数) in the following month. The better CSI 300 index performs in this period, the **less** this WMP pays its return.
- “薪加薪16号” WMP issued by GDB (广发银行) provides a random annualized rate of return, which is also determined by the performance of USD/HKD rate in the following 3 months.

More Complicated: Path-dependent WMP

A WMP issued by CMB (招商银行) has a random annualized rate of return and **random** maturity (到期日) determined by the performance of CSI 500 index (中证500指数), according to the following rules:

- Set the **closing quote** (收盘报价) of CSI 500 index on Feb. 15, 2022 as the reference point (基准点). Monitor the closing quote of CSI 500 index for every 45 days to observe whether it is higher than the reference point. Monitor for at most 8 times.
- If the closing quote of CSI 500 index is higher than the reference point on a monitoring day, then the WMP immediately expires with annualized rate of return 6%.
- If the closing quotes of CSI 500 index are lower than the reference point for all the 8 monitoring days, then the WMP expires on the 8th monitoring day with annualized rate of return 2%.

Financial Derivative

- A **financial derivative** contract is a financial asset whose payoff depends on the value of some **underlying variables**.
- The underlying variable can be a traded asset (e.g., Guizhou Maotai stock), an index (e.g., CSI 300 Index), other derivative (e.g., crude oil futures), interest rates, FX rates, and so on.
- Common financial derivatives
 - futures (期货)
 - forward (远期)
 - option (期权)
 - swap (互换)

Market of Financial Derivatives

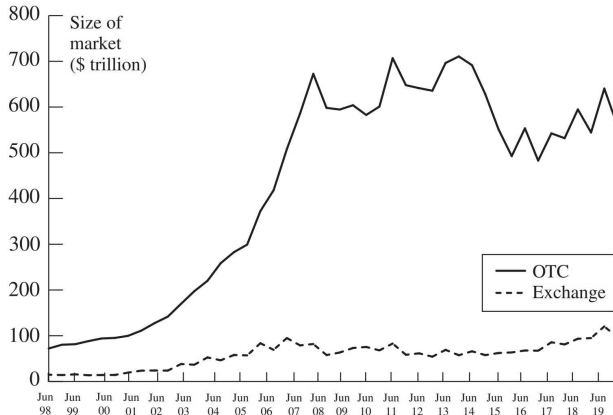
- Non-standard financial derivative has a long history, dating back to the ancient Greece.
- Modern financial derivative first appeared in CBOT (芝加哥商品交易所) founded in 1848. CBOT led the standardizations of cereal commodities, and then futures. Nowadays CBOT becomes a part of CME.
- Standard options first appeared in CBOE (芝加哥期权交易所) founded in 1973.
- Market makers (做市商) provide the liquidity in the trading market.

The OTC Market

- Not all the trading activities occur in the exchanges.
The over-the-counter market (场外市场, OTC market) is important substitute of exchange market.
- Decentralized (去中心化) system. Signed by the delegates of the two parties.
- No standardization. The payoff structure can be complicated and personalized.
- Example: exotic options, introduced in Chapter 2.

Volumes of the OTC market

Size of over-the-counter and exchange-traded derivatives markets



Source: the bank for international settlements

Financial Derivatives in Mainland China

- Forwards and swaps since 2005
- Securities margin trading (融资融券交易) since 2010, and the 7th batch of underlying stocks in 2022.
- Options since 2015
- Commodity futures market, largest in the world.

Special Events

- First option: SSE 50 ETF option (上证50ETF期权) issue on Feb 9, 2015 by SSE (上海证券交易所)
- Commodity options issued in 2017 – American options
- Crude oil futures issued in 2018
- CFFEX (中国金融期货交易所): CSI 300 index option issued in 2019
- Guangzhou Futures Exchange, founded in 2021
- Chinext Board (创业板) ETF option and CSI 1000 index option, 2022

Futures

- A **futures** contract is a financial agreement between two parties to buy or sell the underlying asset at a certain time in the future for a certain price.
- Commodity futures and financial futures
- Long position (to buy) and short position (to sell)
- In comparison, a spot contract makes the transaction happen immediately (or very soon).
- Contract name (in mainland China):
Abbreviation of the variety (品种) + yy + mm
For example: au1812, rb2010, j2509.

To “Short” Something...

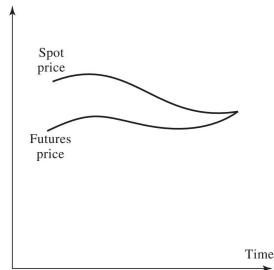
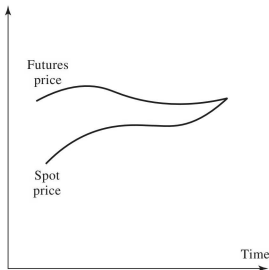
- Many financial assets allow the investor to “short” (卖空) them – one can sell an asset when she does not hold this asset, and buy this asset in future to “close the position” (平仓).
- Imagine that your holding position (持仓量) is negative.
- Default (违约) risk
- The futures market naturally allows short positions.

Example

- On March 2026, you open a short position of 2 lots (手) of the gold futures contract au2606 in SHFE (上海期货交易所), at a price of 588 Yuan per gram.
- One lot of gold futures is 1000 grams (the multiplier).
- **Quiz 1.** If the price of au2606 becomes 560 Yuan per gram on June 2026, then how much is your profit?
- **Quiz 2.** If the price of au2606 becomes 600 Yuan per gram on June 2026, then how much is your loss?

Convergence of Futures and Spot Prices

- **Basis (基差)** is spot price minus futures price. The basis exists because of the interest rate (利率), inventory costs, financing costs (融资成本), etc.
- The basis converges (收敛) to 0 as the maturity approaches.



Margin Account and Daily Settlement

- The futures exchange (交易所) sets margin accounts (保证金账户) to avoid the investors' default risk.
- The **margin (保证金)** is an amount of money frozen (冻结) in the margin account.
- A futures contract is in effect closed out and rewritten at a new price each day.
- The margin account are adjusted to reflect the traders' gain or loss. This practice is referred to as **daily settlement (每日结算)** or **marking to market**.

An Artificial Example

- You long 2 lots of XX futures at a price of \$600/ton. Each lot is 100 tons of XX commodity.
- The initial margin (初始保证金) of each lot of XX futures is \$2000.
- Whenever your account balance (账户余额) is less than the “maintenance margin” (维持保证金), i.e., 75% of the initial margin, then you will receive margin call (催讨通知).
- At that time, if you do not add the margin, the exchange would “forced liquidate” (强制平仓) your position.

Details in the Margin Account

Day	Futures price (\$)	Daily gain (loss) (\$)	Cumulative gain (loss) (\$)	Margin account balance (\$)	Margin call (\$)
	600.00			4,000	
June 5	597.00	(600)	(600)	3,400	
June 6	596.10	(180)	(780)	3,220	
June 9	598.20	420	(360)	3,640	
June 10	597.10	(220)	(580)	3,420	
June 11	596.70	(80)	(660)	3,340	
June 12	595.40	(260)	(920)	3,080	
June 13	593.30	(420)	(1,340)	2,660	1,340
June 16	593.60	60	(1,280)	4,060	
June 17	591.80	(360)	(1,640)	3,700	
June 18	592.70	180	(1,460)	3,880	
June 19	587.00	(1,140)	(2,600)	2,740	1,260
June 20	587.00	0	(2,600)	4,000	
June 23	588.10	220	(2,380)	4,220	
June 24	588.70	120	(2,260)	4,340	
June 25	591.00	460	(1,800)	4,800	
June 26	592.30	260	(1,540)	5,060	

Leverage Effect

- The futures market allows the investor to obtain leverage.
- With a relatively small initial margin, a large speculative position can be taken.

$$\text{leverage ratio (杠杆率)} \times \text{margin rate (保证金率)} = 1.$$

- Leveraged trading takes the risk of forced liquidation and even blowing up (穿仓).
- Various sources of the leverage effect (杠杆效应)

Case: “Negative Equity” in HK

- Real estate prices in Hong Kong keep rising for decades, but dropped much in 2003 and 2008.
- Minimum down payment (首付) ratio is 10% in HK.
- A customer purchased a 30.3 m² apartment at Sha Tin with 6.36 million HKD (港币).
- Its price dropped to 5.30 million HKD in 2020 due to the pandemic. It became “negative equity” if the down payment ratio was 10%.
- There are around 1,000 cases of negative equity in HK in 2022. The financial system faces challenges.

Case: Fu's Tragedy

- FU, Xiaojun (付晓军), an expert on rubber industry and investment with more than 20 years' experience, suicides at Lujiazui on Sept. 28, 2017.
- He held long positions of rubbers on both spot market and futures market, and lost 110 million Yuan in only two weeks.

日期	客户权益 (万)	入金 (万)	当日亏损 (万)	可用资金 (万)	风险率
2017/9/26	3210	0	-6.8	-4.4	100%
2017/9/25	3217	0	-1491	79	98%
2017/9/22	4710	1000	-1934	-1908	140%
2017/9/21	5645	0	-506	-2968	152%
2017/9/20	6151	510	-555	-2548	141%
2017/9/19	6196	0	-484	-3158	150%
2017/9/18	6600	2500	-3234	-2755	141%
2017/9/15	7415	0	-1618	-3866	152%
2017/9/14	9033	0	-1119	-2523	127%
2017/9/13	10152	0	-132	-1594	157%

Rubber Price in 2017

Daily prices of rubber futures from Dec. 15, 2016 to Oct. 16, 2017



Futures Market in China

- SHFE (上海期货交易所): crude oil, copper, rubber, steal bars, gold...
- DCE (大连商品交易所): soybean, egg, corn, plastics, PVC, iron ore...
- ZCE (郑州商品交易所): cotton, sugar, apple, methanol, glass, thermal coal...
- GFE (广州期货交易所): silicon
- CFFEX (中国金融期货交易所): stock index, treasury bond (国债)

Apple Futures and Targeted Poverty Alleviation

- Apple futures is issued by ZCE in 2017. This is the first futures variety of fresh fruit all over the world.
- Standard: 1st class, Red Fuji, diameter $\geq 80\text{mm}$, hardness $\geq 6.5\text{kgf/cm}^2$.
- China produces over a half apples on the world. Many apple production areas overlap with poverty areas. Trading apple futures can help predict the apple prices in future and adjust the production plan. This is a part of the targeted poverty alleviation (精准扶贫战略) implemented by Chinese Government.

Bonus Scene (彩蛋): Negative Futures Price

- On Apr. 20, 2020, NYMEX (纽约商业交易所) crude oil futures price dropped about 300% and closed at \$-37.63/barrel. The lowest price is \$-40.32/barrel.

US oil prices turn negative

Price per barrel of WTI



- D. Trump: very interesting to a lot of people...

Forwards

- The definition of a **forward** contract is completely the same with futures contract – a financial agreement between two parties to buy or sell the underlying asset at a certain time T in the future for a certain price K .
- Of course, they have apparent differences:
 - Forwards are only traded in the OTC market
 - No daily settlement
 - Underlying assets of forwards are all interest rate or FX rate.

Comparison between Forwards and Futures

Comparison of forward and futures contracts

Forward

Private contract between two parties
Not standardized
Usually one specified delivery date
Settled at end of contract
Delivery or final cash settlement usually takes place
Some credit risk

Futures

Traded on an exchange
Standardized contract
Range of delivery dates
Settled daily
Contract is usually closed out prior to maturity
Virtually no credit risk

Payoff

- The **payoff (支付)** from a long position in a forward contract is

$$S_T - K$$

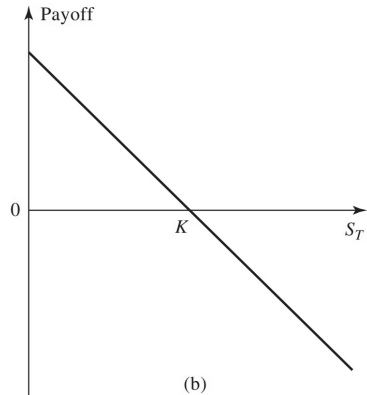
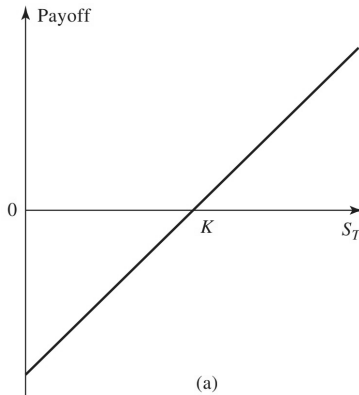
where K is the strike (delivery price, exercise price, 执行价) and S_T is the spot price of the underlying asset S at maturity T .

- Reason: the holder of the contract is obligated to **buy an asset worth S_T for K** .
- The payoff from a short position in a forward contract:

$$K - S_T.$$

The Payoff Functions

Payoffs from forward contracts: (a) long position, (b) short position.
Delivery price = K ; price of asset at contract maturity = S_T .



Arbitrage

- An **arbitrage** (套利) is a transaction that involves no negative cash flow at any state and a positive cash flow in at least one state.
- In simple terms, it is the possibility of a risk-free profit.
- Quantitatively, an arbitrage portfolio (投资组合) has a random profit X such that

$$P(X < 0) = 0 \quad \text{and} \quad P(X > 0) > 0.$$

No-arbitrage principle

- **No-arbitrage principle (无套利原则)**: market prices of all the assets do not allow any arbitrage opportunity.
- **Law of one price (LOOP, 一价律)**: If two portfolios X and Y have the same terminal payoff, then their initial costs are the same as well. Namely,

$$X_T \equiv Y_T \quad \Rightarrow \quad X_t = Y_t.$$

- The law of one price is a corollary (推论) of no-arbitrage principle. (Why?)

Construct Arbitrage Portfolios...

- A forward contract F has an underlying asset S . Define two **stochastic processes** (defined in Chap. 2) $\{F_t\}$ and $\{S_t\}$ to denote the price processes.
- Consider the following two portfolios at time 0:
 - (X) Long 1 share of S with a cost S_0 ;
 - (Y) Long 1 share of F with a cost F_0 , and a cash deposit (现金存款) of Ke^{-rT} .
- Here, r is the **continuous interest rate**.
 - Cash deposit of D becomes De^{rT} after a period of T .

Forward Pricing

- We compute

$$Y_T = (S_T - K) + Ke^{-rT}e^{rT} = S_T = X_T,$$

and then conclude

$$S_0 = F_0 + Ke^{-rT}$$

based on the law of one price.

- F_t : the value of forward contract F at time t
- **Forward pricing**: in the real marketplace, we always set $K = S_0e^{rT}$ to **eliminate** the initial value F_0 .

European Options

- Options are traded both on exchanges and in the OTC market.
- A **call option** (看涨期权) gives the holder the right to buy the underlying asset on a date T for a price K . A **put option** (看跌期权) gives the holder the right to sell the underlying asset on T for K .
- They are called European options or vanilla options.
- K : exercise price or strike price
 T : expiration date or maturity.

Option Pricing

- It should be emphasized that an option gives the holder the right to do something. The holder **does not have to** exercise this right.
- This is what distinguishes options from forwards and futures, where the holder is **obligated** to buy or sell the underlying asset. Therefore, it is impossible to cost nothing to enter into an option contract.
- The aim of the **option pricing theory** (期权定价理论), introduced later in Chapter 2, is to find a “fair” initial value of an option contract.
- **Quiz 3.** What is the main difference between forward pricing and option pricing?

Example: Option Quotes

ETF期权		510050 50ETF		最新 2.795 幅度% -0.72 涨跌 -0.020 买入 2.754 卖出 2.755 持仓 20 总量 252.23万 成交 6.97亿 换手 一 量比 0.94		融资融券												
50ETF期权4月2800		认购										认沽						
<< 买入	卖出	最新	涨跌	幅度%	理论价	隐含%	总量	持仓	执行价	买入	卖出	最新	涨跌	幅度%	理论价	隐含%	总量	持仓 >>
0.2632	0.2634	0.2616	-0.0190	-6.77	0.2599	28.60	654	2579	2.500	0.0037	0.0038	0.0038	-0.0001	-2.56	0.0006	31.61	6366	42920
0.2140	0.2153	0.2120	-0.0159	-6.55	0.2113	25.84	2560	6804	2.550	0.0058	0.0059	0.0058	0.0000	0.00	0.0019	29.24	6713	41022
0.1687	0.1700	0.1680	-0.0182	-9.77	0.1647	26.39	3496	8774	2.600	0.0100	0.0101	0.0101	0.0002	2.82	0.0052	27.82	8778	35337
0.1264	0.1276	0.1263	-0.0150	-11.12	0.1218	26.00	8467	19511	2.650	0.0175	0.0176	0.0177	0.0020	12.74	0.0122	26.71	15490	47903
0.0886	0.0890	0.0890	-0.0141	-13.68	0.0845	25.37	17718	33699	2.700	0.0296	0.0297	0.0297	0.0043	16.93	0.0248	25.56	26646	51469
0.0581	0.0582	0.0581	-0.0101	-14.81	0.0544	24.71	47051	56774	2.750	0.0485	0.0489	0.0485	0.0082	20.35	0.0447	24.77	37395	51442
0.0353	0.0357	0.0354	-0.0071	-16.71	0.0323	24.48	37904	79255	2.800	0.0757	0.0758	0.0759	0.0121	18.87	0.0725	24.64	30688	33213
0.0200	0.0202	0.0202	-0.0045	-18.22	0.0176	24.52	22460	80110	2.850	0.1098	0.1100	0.1112	0.0144	14.88	0.1077	25.04	12618	25072
0.0108	0.0110	0.0110	-0.0025	-18.52	0.0087	24.81	11912	67042	2.900	0.1516	0.1520	0.1518	0.0167	12.36	0.1487	25.42	5472	16832
0.0055	0.0056	0.0055	-0.0016	-22.54	0.0039	24.93	8911	47409	2.950	0.1953	0.1966	0.1969	0.0196	11.05	0.1938	26.53	2764	7040
0.0030	0.0031	0.0030	-0.0012	-28.57	0.0016	25.83	6299	54412	3.000	0.2425	0.2440	0.2449	0.0199	8.84	0.2414	28.94	477	3833
0.0014	0.0015	0.0014	-0.0003	-17.65	0.0002	29.55	1721	41036	3.100	0.3481	0.3419	0.3437	0.0187	5.75	0.3399	35.63	1217	9915
0.0009	0.0010	0.0010	0.0000	0.00	0.0000	34.47	485	42931	3.200	0.4397	0.4415	0.4422	0.0172	4.05	0.4395	40.20	179	10856

Option's Payoffs

- The payoff function of the long position of a European call option is

$$C(T, K) := \begin{cases} S_T - K, & \text{if } S_T \geq K; \\ 0, & \text{if } S_T < K, \end{cases}$$

or equivalently, denoted by the sign of “the positive part”:

$$C(T, K) := (S_T - K)^+. \quad (\text{Why?})$$

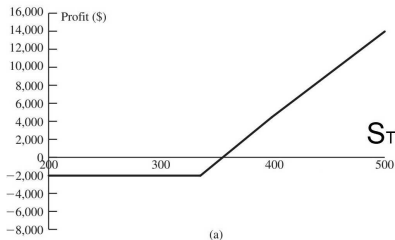
- Similarly, payoff function of the long position of European put option is

$$P(T, K) := (K - S_T)^+. \quad (\text{Why?})$$

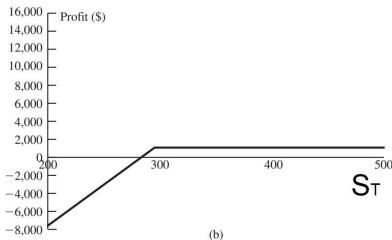
The P&L Chart of Options

- P&L (profit and loss, net profit, 损益) is the revenue minus the cost.
- P&L of the long position of an option is the terminal payoff minus the initial option price (if we ignore the interest).

Call Option P&L (Long)



Put Option P&L (Short)



Leverage Effect of Options

- Option contracts naturally contain the leverage effect.
- Example: the price of a call option with $S_0 = 100$, $K = 100$ and $T = 1$ typically ranges from 6 to 9.
- If S_t rises a lot, the return of the call option can be over 10 times larger than the return of S .

$$R_S = \frac{200 - 100}{100} \quad \text{vs.} \quad R_C = \frac{200 - 100 - 7}{7}$$

- Short selling is dangerous.

Case: Bankruptcy of CNAF (SG)

- CNAF (SG) (中航油新加坡公司) started its business on crude oil derivative hedging on 2003, to manage the risk in oil trading. CEO CHEN, Jiulin (陈九霖) privately expanded the business to option speculation, and shorted call options of crude oil.
- Margin call: whenever the oil price increased by 1 USD, the company had to add 50 million USD.
- On Oct. 26, 2004, the positions forced liquidated lost 394 million USD, and the positions to be forced liquidated would lose other 160 million USD.
- CNAF (SG) went bankrupt on Dec. 2004. Chen was sentenced to 4 years 3 months in prison.